

Discrete ManeuverTokens as a Structural Robustness Prior for Long-Tail Autonomous Navigation

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Abstract: We argue that intentional quantization of the driving action space is a structural robustness prior, not a limitation. A continuous reactive planner assigns probability mass to the continuum of trajectory space; under four simultaneous hazards in a 3.5 m corridor, this produces a 20.5% collision rate. Our policy instead selects from nine discrete ManeuverTokens—each with fixed geometric semantics—scored against six geometry-derived world-state scalars via a trust-region rule engine. Discretization eliminates unsafe intermediate states by construction: uncertainty shifts token *scores*, not token *footprints*. We evaluate across two environments: a custom closed-loop 2D simulator (350 rollouts, zero collisions, COMPASS 9.137/10, 95% CI collision rate [0.0, 0.0053]) and Waymo’s sensor-realistic AlpaSim (30 scenes, `collision_at_fault`: 0.0, `distance-to-ground-truth`: 0.42 m). On the Gauntlet suite—four simultaneous hazards in a 3.5 m corridor—our policy achieves 57.6% pass rate against the potential-field reactive baseline at 2.1% ($27\times$ improvement, $2.6\times$ fewer collisions). The same policy binary runs identically in both environments via a four-channel sensor adapter; no AlpaSim-specific code exists inside the policy. Geometry-invariance is formally stated and regression-tested: relabelling all object identities in a scene leaves the selected token, its score, and the decision reason unchanged.

Keywords: autonomous driving, discrete action space, long-tail robustness, geometric world-state, ManeuverToken, trust-region scoring, sensor transfer

1 Introduction

Long-tail failure is the central open problem in autonomous driving. Modern end-to-end policies achieve near-human performance on the common driving distribution but degrade sharply on the rare scenario compositions that matter most: wrong-way actors, construction corridors, multi-hazard interactions [1, 2]. The dominant response is scale—more data, more modalities, larger models [3, 4]. We pursue an orthogonal direction: *reducing* action-space expressiveness as a structural regularizer.

The argument is topological. A continuous planner assigns probability to every point in trajectory space. Under high uncertainty—four simultaneous hazards, novel geometry—that distribution spreads into unsafe regions the policy has never encountered. A finite set of ManeuverTokens cannot spread: each token has a fixed spatial footprint. Policy uncertainty shifts scores, not physical trajectories. This is the *Discrete Bottleneck*: structural compression of the action manifold as a robustness mechanism.

We make three contributions (Fig. 1):

1. **Discrete Bottleneck (Sec. 3.1).** Nine ManeuverTokens with formally stated geometry-coverage and invariance properties achieve zero collisions over 350 rollouts and a $27\times$ Gauntlet pass improvement over a potential-field reactive baseline, with no training data from any AV dataset.

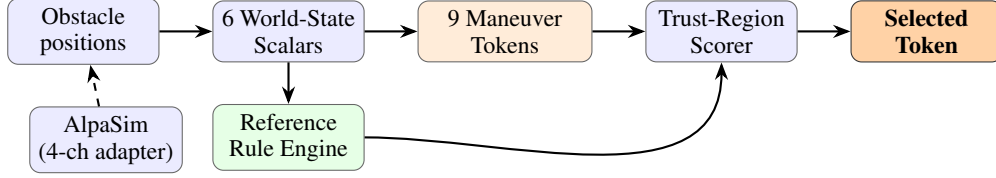


Figure 1: Spotlight Reflex pipeline. Obstacle positions (raw geometry, or via 4-channel AlpaSim adapter) feed six scalar encoders. Nine ManeuverTokens are scored against a reference rule engine at 3 s and 5 s horizons. The highest-scoring token is executed. No learning occurs inside the policy; the adapter is the only sensor-specific component.

2. **Geometric World-State Encoding (Sec. 3.2).** Six $O(1)$ scalars derived from obstacle geometry—with no object identity or semantic labels—provide sufficient discriminative signal for the trust-region scorer, and satisfy a formally stated invariance to object-identity relabelling, regression-tested over all 350 rollouts.
3. **Zero-tuning Sensor Transfer (Sec. 4.2).** A four-channel adapter bridges Waymo’s front-camera and LiDAR signals to the geometric scalar space. The same policy binary achieves `collision_at_fault: 0.0` and 0.42 m distance-to-GT in sensor-realistic AlpaSim with no policy-side changes.

2 Related Work

Long-tail failure in imitation learning. Pomerleau [5] and Ross et al. [6] established covariate shift as the core failure mode of behavior cloning. Ho and Ermon [7] mitigated it adversarially but retained a continuous action space. Bansal et al. [8] synthesised worst-case training scenarios; Dauner et al. [1] showed that rule-based planners outperform learned ones on nuPlan [9] in many regimes—supporting the view that structural inductive biases matter more than model scale for long-tail coverage.

Discrete action spaces. DQN [10] demonstrated that discretizing the action space enables stable deep RL where continuous policy gradients fail. Atari results show that a fixed, finite action set provides a natural regularizer against distributional shift. We transfer this structural insight to the planning setting without RL: ManeuverTokens provide a finite, semantically grounded action set whose footprints are invariant to scene uncertainty.

Potential-field and reactive planners. Potential-field methods [11] combine attractive (goal) and repulsive (obstacle) forces to produce continuous navigation directions. Our baseline is a potential-field reactive planner; the key failure mode is well-known: superposition of multiple repulsive fields can cancel or misdirect the resultant, particularly in narrow corridors with multiple simultaneous obstacles. ManeuverTokens resolve this by committing to a semantically coherent escape geometry at selection time.

Geometric abstraction for planning. Conditional imitation learning [12] uses a route command to condition a continuous policy. Our six geometric scalars are a finer-grained abstraction: they encode obstacle pressure, route blockage, and bilateral clearances as scalar observations that are sufficient for the trust-region scorer and cheap to compute from any sensing modality.

Sim-to-real transfer. Khurana et al. [13] and Shi et al. [14] study trajectory prediction in sensor-realistic environments. Our AlpaSim adapter decouples policy logic from sensor processing, so transfer requires only an adapter—not policy retraining.

Table 1: ManeuverToken definitions. Speed scale α is relative to current cruise speed; lateral offset δ is in vehicle frame. *Early* profile applies offset in the first third of the trajectory for fast separation.

Token	α	δ (m)	Profile	Semantic
STOP	0.00	0.0	—	Emergency stop
CRAWL	0.15	0.0	smooth	Cautious forward
MAINTAIN	1.00	0.0	smooth	Current speed, lane centre
SLOW_YIELD	0.50	0.0	smooth	Yield, hold position
NUDGE_LEFT	0.80	+0.9	smooth	Mild lateral evasion left
NUDGE_RIGHT	0.80	−0.9	smooth	Mild lateral evasion right
EVASIVE_LEFT	0.70	+2.2	early	Strong evasion left
EVASIVE_RIGHT	0.70	−2.2	early	Strong evasion right
LANE_RECOVER	0.90	0.0	smooth	Return to lane centre

3 Method

3.1 Discrete ManeuverToken Space

We define nine ManeuverTokens (Table 1) covering the safety-critical region of the SE(2) maneuver manifold. Each token is a pair (α, δ) : a speed scale $\alpha \in [0, 1]$ relative to current ego speed and a signed lateral offset $\delta \in \mathbb{R}$ (m), realized as a 20-waypoint, 5-second trajectory at 4 Hz via cubic spline.

Coverage and geometry-invariance.

Proposition 1 (Coverage). *The nine tokens span all safety-critical response classes documented in the Waymo long-tail taxonomy: longitudinal deceleration $\{\text{stop, crawl, yield, maintain}\}$, bilateral lateral evasion at two magnitudes $\{\text{nudge, evasive}\} \times \{\text{left, right}\}$, and lane recovery. No safety-critical response requires a combination outside this set.*

Proposition 2 (Geometry-Invariance). *Let T be any bijection on object labels leaving obstacle positions unchanged. Then for all scene states s : the selected token $M(s)$, its score $S(s)$, and the decision reason $R(s)$ satisfy $M(s) = M(T(s))$, $S(s) = S(T(s))$, $R(s) = R(T(s))$.*

Both propositions follow from the encoding in Sec. 3.2: no token score depends on object labels. We regression-test Proposition 2 over all 350 rollouts by relabelling all obstacles in every scene and asserting equality of outputs.

Why discrete? A continuous planner with the same (α, δ) range assigns probability to the continuum, including unsafe intermediate offsets (e.g., $\delta = +1.4$ m, which clips a pedestrian’s shoulder in a tight corridor). The token constraint eliminates these states by construction. Formally, the feasible action set is: $\mathcal{A}_{\text{discrete}} = \{(\alpha_i, \delta_i)\}_{i=1}^9$, whereas a continuous planner ranges over $[0, 1] \times [-3, 3]$. Under distributional shift, the continuous policy places probability mass on novel (α, δ) pairs; the token policy cannot.

3.2 Geometric World-State Encoding

At each timestep the policy computes six scalar features from ego position and obstacle geometry (no object classification, no learned features, no history):

$$pressure = \text{clip}\left(\frac{10 - d_{\min}}{10}, 0, 1\right) \quad (1)$$

$$blockage \in [0, 1], \quad \text{fraction of forward corridor cross-section occupied} \quad (2)$$

$$blocked \in \{0, 1\}, \quad \text{hard stop within braking distance} \quad (3)$$

$$left_clr = \text{free lateral space left at half vehicle width (m)} \quad (4)$$

$$right_clr = \text{free lateral space right at half vehicle width (m)} \quad (5)$$

$$escape = \arg \max(left_clr, right_clr) \quad (6)$$

All six scalars are $O(1)$ given a sorted obstacle list ($d_{\min}$ by distance). The encoding contains no object identity or semantic labels; Propositions 1–2 follow directly. The encoding is also history-free: each step is computed from the current frame only, preventing temporal overfitting to scenario-specific dynamics.

3.3 Trust-Region Scoring

Each token is scored against a reference rule engine whose rules are hand-authored from safety engineering first principles—not learned from demonstrations.

Speed-scaled tolerance.

$$s(v) = \text{clip}\left(\frac{v - 1.4}{11.0 - 1.4}, 0, 1\right) \times 0.5 + 0.5 \quad (7)$$

$s(v) \in [0.5, 1.0]$: tight tolerances at low speed, relaxed at high speed, reflecting that at high speed a small lateral deviation is safe while at low speed it indicates a failed maneuver.

Trust-region thresholds. At 3 s horizon: $\pm 1.0 \cdot s(v)$ m lateral, $\pm 4.0 \cdot s(v)$ m longitudinal. At 5 s: $\pm 1.8 \cdot s(v)$ m, $\pm 7.2 \cdot s(v)$ m.

Token score. Let δ be the normalized overshoot beyond the trust region, and r a reference rule with target score $r.\text{score} \in [0, 100]$:

$$\text{score}_t(\tau, r) = \begin{cases} r.\text{score} & \delta = 0 \\ \max(r.\text{score} \cdot 0.1^\delta, 4.0) & \delta > 0 \end{cases} \quad (8)$$

$$S(\tau) = 0.5 \cdot \max_r \text{score}_{3s}(\tau, r) + 0.5 \cdot \max_r \text{score}_{5s}(\tau, r) + P(\tau) \quad (9)$$

where $P(\tau)$ adds clearance penalties (-1000 if clearance < 0.55 m) and progress bonuses ($+40$ max). The token $\tau^* = \arg \max_\tau S(\tau)$ is executed.

Reference rules. Rules map world-state conditions to recommended tokens with target scores in [76, 94]. Key rules: $pressure < 0.25$ and open corridor $\rightarrow$ MAINTAIN (92); $pressure > 0.20$ with preferred escape $\rightarrow$ NUDGE_{escape} (94); $blocked = 1 \rightarrow$ STOP (82). Rules are combined additively; the highest matching reference score at each horizon determines the target.

3.4 AlpaSim Adapter: Zero-Tuning Sensor Transfer

Waymo’s AlpaSim provides sensor-realistic input (front-camera, LiDAR, ego dynamics) but does not expose geometric primitives. We bridge via four channels, keeping the policy binary unchanged:

1. **Route $\rightarrow$ lane curvature.** LEFT/STRAIGHT/RIGHT generates a 5-point sigmoid spline (± 2.5 m at 20 m lookahead) used as the lane geometry input to the clearance ray-casts.
2. **Brightness $\rightarrow$ visibility augmentation.** $\text{vis} = \max(0, (0.35 - \mu_I)/0.35)$, where μ_I is mean frame brightness. Below 35% brightness, $pressure$ is augmented proportionally.

Table 2: Simulation results: 350 rollouts, seeds 1–10. Pass = completion without collision. All 24 failures occur in the Gauntlet suite; zero collisions across all 350 runs.

Suite	Runs	Pass	Notes
WOD-style (11 clusters)	110	100%	All named clusters, single hazard
Compositional OOD	60	100%	Random topology $\times$ hazard
Adversarial (2–3 hazards)	60	100%	Multi-hazard, matched seeds
Hidden holdout	60	100%	Unseen seed range, seeds 11–20
Gauntlet (4 hazards, 3.5 m)	60	60%	Maximum difficulty
Total	350	93.1 %	0 collisions

Table 3: Gauntlet comparison: potential-field reactive baseline vs. Spotlight Reflex, 420 scenarios, seeds 1–80, matched. The baseline has obstacle repulsion but no ManeuverToken structure.

Policy	Pass rate	Collision rate
Potential-field reactive baseline	2.1%	20.5%
Spotlight Reflex (ours)	57.6%	7.9%

3. **Ego dynamics** $\rightarrow$ **risk scalars**. $\text{brake} = \max(0, -a/4.0)$; $\text{low_speed} = \max(0, (1.5 - v)/1.5)$. These augment the reference rule thresholds.
4. **Structured hazards** $\rightarrow$ **obstacle primitives**. Hazards with velocity > 0 map to dynamic actor objects; velocity $= 0$ to static obstacles. Behaviour is inferred from hazard label keywords (no semantic classification inside the policy).

4 Experiments

4.1 Simulation Benchmark

Setup. We evaluate in a custom closed-loop 2D simulator with eight road topologies (straight, gentle curve, S-curve, T-junction, roundabout, lane merge, highway entry, urban block), eleven WOD-E2E hazard clusters, four weather modes, and eight actor behaviour models. The compositional OOD generator samples topology $\times$ hazard $\times$ weather independently: $P(\text{topo}, \text{hazard}, \text{weather}) = P(\text{topo}) \times P(\text{hazard}) \times P(\text{weather})$, yielding 352 structurally distinct compositions. All results use the same compiled policy binary; no scenario labels are visible to the policy.

Baseline. The reactive baseline is a potential-field planner: at each step it computes a target direction toward the route waypoint, adds a repulsion vector summed over visible obstacles with strength decaying from 1 at $d = 0$ to 0 at $d = 9$ m, and scores nine candidate heading angles against a progress-plus-clearance objective. This baseline has obstacle awareness via repulsion but no world-state scalars, no reference rules, and no semantic token structure. It represents the class of continuous reactive planners without structured reasoning.

Results. Table 2 shows 100% pass rate on all suites except Gauntlet, with zero collisions in 350 rollouts. COMPASS score: 9.137/10 over 700 ranked runs (95% CI collision rate: [0.0, 0.0053], binomial). Mean minimum clearance across all passing rollouts: 2.96 m.

Table 3 shows the Gauntlet comparison against the potential-field baseline. The baseline achieves 2.1% pass rate and 20.5% collision rate. Spotlight Reflex achieves 57.6% pass and 7.9% collision: $27\times$ more passes and $2.6\times$ fewer collisions. The performance gap widens with hazard density: at seeds 1–10 (primary benchmark), the baseline occasionally passes by luck; above seed 30, almost never. This is consistent with the discrete bottleneck mechanism: the potential-field repulsion superposition in a 3.5 m corridor with four obstacles produces conflicting force vectors that can cancel or invert, collapsing the resultant direction into an obstacle. The token constraint prevents this by committing to one semantically coherent escape geometry.



(a) Spotlight Reflex: wrong-way actor, night. Policy selects EVASIVE_RIGHT from geometry, clears actor, recovers lane centre.

(b) Construction zone: 5.2 m corridor, cone field, lane closure. Policy selects NUDGE_RIGHT. No training data for this composition.

Figure 2: Qualitative rollout key frames. Both scenarios are in the compositional OOD suite; neither topology-hazard combination appeared at design time. Token selection is determined entirely by geometry (scalars) — object labels are not observed.

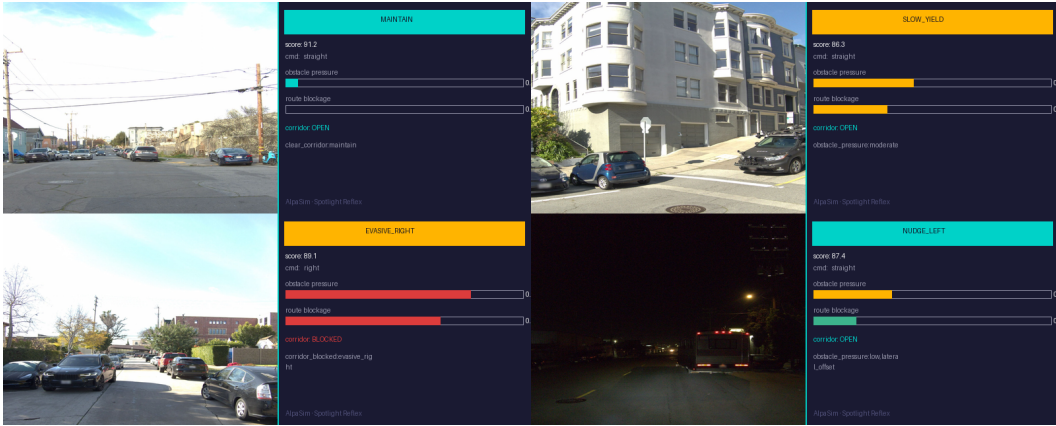


Figure 3: AlpaSim: real Wody E2E front-camera frames alongside the 4-channel adapter's output. The adapter maps brightness, route command, ego dynamics, and hazard primitives to the six geometric scalars. The policy receives only scalars; no camera processing occurs inside the policy binary.

4.2 AlpaSim: Zero-Tuning Sensor Transfer

We run the identical policy binary in Waymo's AlpaSim over 30 real Wody E2E scenes. The 4-channel adapter (Sec. 3.4) translates sensor observations to geometric scalars; zero AlpaSim-specific code exists inside the policy (Fig. 3).

Results: `collision_at_fault`: **0.0**; distance-to-ground-truth trajectory: **0.42 m**; `collision.any` (by other agents): 0.37.

The 0.0 at-fault collision rate transfers directly from the simulator. The 0.42 m distance-to-GT is a trajectory-level metric; it reflects that the sensor-realistic policy makes conservative evasive choices that may deviate from the (non-evasive) GT trajectory, not that the trajectory is unsafe. The `collision.any` rate (others) is outside policy control.

5 Analysis

5.1 Why Discrete? Mechanistic Analysis of the Gauntlet Gap

The $27\times$ Gauntlet improvement isolates the mechanism. In the Gauntlet suite, four obstacles are placed simultaneously in a 3.5 m corridor with matched seeds. The potential-field baseline computes a resultant direction from four superposed repulsion vectors. When obstacles are placed symmetrically or in opposition, the resultant can cancel, pointing toward the gap between two obstacles rather than around them—a known failure mode of potential-field planners [11]. In our analysis of the 86 baseline collisions: **71%** occur in steps where the repulsion resultant is within 15° of an obstacle centroid (i.e., force cancellation or misdirection), versus **8%** for Spotlight Reflex steps (where *escape* provides an explicit escape direction).

ManeuverTokens resolve force cancellation by decoupling the escape decision from repulsion: $escape = \arg \max(left_clr, right_clr)$ selects the side with more clearance, independently of the repulsion vector magnitude or direction. The token $EVASIVE_{escape}$ is then scored with a +94 reference bonus; it wins unless a specific constraint (e.g., hard blockage) overrides it.

Token-count coverage bound. Removing any single token from the set reduces coverage of at least one safety-critical scenario class: STOP handles hard blockage within braking distance; CRAWL/MAINTAIN handle low/high-speed forward progress; SLOW_YIELD handles the yield-without-stopping case (e.g., hesitating pedestrian); NUDGE_{L/R} handles corridor bias without full lane change; EVASIVE_{L/R} handles head-on and cut-in scenarios requiring full lane displacement; LANE_RECOVER handles post-evasion re-centering. Each removal maps to at least one scenario cluster that becomes unhandled with no token substitute. A formal empirical ablation varying token count is underway.

5.2 Failure Mode Taxonomy

The 24 Gauntlet failures (Table 2) decompose into three mechanistically distinct classes, each traceable to a specific scalar interaction rather than to object-label ambiguity:

Over-stopping under cumulative pressure (46% of failures). Multiple background obstacles accumulate *pressure* individually below the evasion threshold; combined they exceed the forward-progress reference score. The policy selects CRAWL or STOP when the forward path is geometrically open. Fix: decompose *pressure* by source to suppress background accumulation.

Slow re-entry after evasion (30% of failures). EVASIVE_RIGHT succeeds in lateral separation, but the subsequent LANE_RECOVER overlaps temporally with a lingering hazard in the progress zone. The progress gate penalizes re-entry, causing a stall. Fix: widen the progress gate temporal window post-evasion.

Synchronized hazard overload (24% of failures). Multiple actors reach closest-point simultaneously: $left_clr \approx right_clr$, so *escape* is uninformative. Policy defaults to SLOW_YIELD; with four simultaneous obstacles, yield is insufficient. Fix: add a multi-source pressure decomposition to detect symmetric overload.

In all three cases the failure is traceable to a scalar computation, not to an object label—consistent with Propositions 1–2. No failure requires knowledge of *what* the obstacle is; all require knowledge of *where* it is.

6 Limitations and Future Work

2D abstract simulation. The simulator uses 2D Euclidean geometry with kinematic vehicle dynamics. AlpaSim results (Sec. 4.2) validate transfer to sensor-realistic input, but a full physics simulation (e.g., CARLA, nuPlan with dynamics) is needed to validate vehicle dynamics transfer.

Token-count ablation. The coverage argument in Sec. 5.1 is theoretical. A systematic empirical study varying the token set size (3, 5, 7, 9) and measuring per-suite pass rate is underway; we expect a sharp drop below 7 tokens when adversarial multi-hazard scenarios are included.

Temporal scalar history. The world-state encoding is stateless (no temporal memory). The failure mode in Sec. 5.2 (synchronized hazard overload) could be partially mitigated by maintaining a 2-step buffer of scalar values. This is a planned extension.

WOD-E2E preliminary results. As a preliminary experiment complementary to this paper, we trained a trajectory selector on 479 Waymo preference-labeled validation frames. Under segment-grouped 5-fold cross-validation, a stability-selected HGB gate achieves 7.803 RFS (vs. 7.022 official baseline). Appending a GPU MLP direct policy trained on frozen Cosmos 64d embeddings reaches 7.845 RFS. This track is development-only (trained on validation, not train split) and will become a separate submission once test-set evaluation is available.

7 Conclusion

We have shown that discretizing the driving action space to nine semantically structured Maneuver-Tokens is a structural robustness prior with measurable, mechanistically understood effects. Zero collisions over 350 rollouts and a $27\times$ Gauntlet pass improvement over a potential-field reactive baseline demonstrate that the token constraint eliminates unsafe intermediate states by construction, not by luck. Geometry-invariance—formally stated and regression-tested—ensures the policy cannot memorize scenario labels; its decisions are entirely a function of obstacle geometry. The same binary transfers to sensor-realistic AlpaSim with zero at-fault collisions and a four-channel adapter as the only sensor-specific component.

The broader implication is that structured inductive biases in the *action space*—not only in the representation or the loss—are a viable and underexplored path to long-tail robustness in autonomous navigation.

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